

General theorems on Hilbert Spaces

based on “Les maths en tête - Analyse”, X. Gourdon

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A real Hilbert space H is a vector space endowed with an **inner product** $\langle \cdot, \cdot \rangle$, which is a symmetric, definite positive bilinear form¹.
The corresponding norm is given by

$$\| \cdot \| : \begin{cases} H \longrightarrow \mathbb{R}_+ \\ x \longmapsto \sqrt{\langle x, x \rangle} \end{cases}$$

Moreover H is **complete** w.r.t. the distance

$$\forall (x, y) \in H^2, \quad d(x, y) := \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$$

Since we are interested in the general case, we will also assume that H is infinite dimensional, e.g. in the last section about Hilbert bases.

For all $(x, y) \in H^2$, one has the

$$\begin{cases} \textbf{Cauchy-Schwarz' ineq.} : & |\langle x, y \rangle| \leq \|x\| \|y\| \\ \textbf{Parallelogram identity} : & \|x + y\|^2 + \|x - y\|^2 = 2 \|x\|^2 + 2 \|y\|^2 \end{cases}$$

¹We will assume for simplicity that H is a real Hilbert, the proofs being similar for spaces over $\mathbb{C}$.

- 1 Hilbert Projection Theorem
- 2 Orthogonal Complement
- 3 Riesz Representation Theorem
- 4 Hilbert/(Orthonormal) Bases

1/ Projection on a Closed Convex Set

Theorem (p. 407 in the 2008 ed.; p. 427 in the 2020 edition)

Let H be a Hilbert space and $C \subset H$ a non-empty closed convex subset. For every $x \in H$, there exists a unique element $x_C \in C$ such that:

$$d(x, C) := \inf_{z \in C} \|x - z\| = \|x - x_C\| \quad (1)$$

x_C is the **orthogonal projection of x onto C** . It is the unique element in C s.t.

$$\forall z \in C, \quad \langle z - x_C, x - x_C \rangle \leq 0 \quad (2)$$

Proof: Existence in (1). Let us denote $\delta := d(x, C)$ and recall the characterization of the infimum (for a subset of $\mathbb{R}$):

$$\forall \epsilon > 0, \quad \exists z \in C, \quad d(x, C) + \epsilon > \|x - z\| \geq d(x, C)$$

So for a sequence $(\epsilon_n)_{n \in \mathbb{N}} \in (\mathbb{R}_+^*)^{\mathbb{N}}$ converging to 0, a repeated use of the previous assertion yields a so-called **minimizing sequence** $(z_n)_{n \in \mathbb{N}} \in C^{\mathbb{N}}$ such that $\lim_{n \rightarrow +\infty} \|x - z_n\| = \delta$. Let us show that $(z_n)_{n \in \mathbb{N}}$ is convergent.

One obtains with the parallelogram identity that for all $p, q \in \mathbb{N}$,

$$\|(x - z_p) + (x - z_q)\|^2 + \|(x - z_p) - (x - z_q)\|^2 = 2 (\|x - z_p\|^2 + \|x - z_q\|^2)$$

$$\iff 4 \left\| x - \frac{z_p + z_q}{2} \right\|^2 + \|z_q - z_p\|^2 = 2 (\|x - z_p\|^2 + \|x - z_q\|^2)$$

Since C is convex, $\frac{z_p + z_q}{2} \in C$ so $4 \left\| x - \frac{z_p + z_q}{2} \right\|^2 \geq 4\delta^2$. Hence

$$\|z_q - z_p\|^2 \leq 2 [(\|x - z_p\|^2 - \delta^2) + (\|x - z_q\|^2 - \delta^2)]$$

Recalling that $(\|x - z_n\|)_{n \in \mathbb{N}}$ converges to δ allows to conclude that $(z_n)_{n \in \mathbb{N}}$ is Cauchy. Since H is complete, $(z_n)_{n \in \mathbb{N}}$ has a limit x_C , and since C is closed, $x_C \in C$.

Unicity in (1): let us give two proofs of this fact. If y and z are two elements in C such that $\|x - y\| = \delta = \|x - z\|$, then one can choose the following minimizing sequence: $u_{2k} := z$, $u_{2k+1} = y$ for all $k \in \mathbb{N}$. By the previous paragraph, this sequence is Cauchy so it converges. By unicity of the limit $z = y$.

Characterization (2) of the Projection

Or more directly, if y and z are two elements in C such that $\|x - y\| = \delta = \|x - z\|$. Since $\frac{y+z}{2} \in C$, one obtains as above that

$$\begin{aligned}\|z - y\|^2 &= 2(\|x - y\|^2 + \|x - z\|^2) - 4\left\|x - \frac{y+z}{2}\right\|^2 \\ &\leq 4\left(\delta^2 - \left\|x - \frac{y+z}{2}\right\|^2\right) \leq 0\end{aligned}$$

Equivalence (1)-(2): $(\Leftarrow)$ if $y \in C$ is such that

$$\forall z \in C, \quad \langle z - y, x - y \rangle \leq 0$$

then

$$\begin{aligned}\|z - x\|^2 &= \|(z - y) - (x - y)\|^2 = \|z - y\|^2 - 2\langle z - y, x - y \rangle + \|x - y\|^2 \\ &\geq \|z - y\|^2 + \|x - y\|^2 \geq \|x - y\|^2\end{aligned}$$

so $y \in C$ is indeed the minimum, $\|x - y\| = d(x, C)$.

$\boxed{\Rightarrow}$ let us show that the orthogonal projection x_C satisfies (2).

$$\forall z \in C, \quad \|x - x_C\| \leq \|x - z\|$$

$$\begin{aligned} \Rightarrow \quad \cancel{\|x - x_C\|^2} &\leq \|x - z\|^2 = \|(x - x_C) - (z - x_C)\|^2 \\ &= \cancel{\|x - x_C\|^2} - 2\langle x - x_C, z - x_C \rangle + \|z - x_C\|^2 \end{aligned}$$

Let us now introduce a parameter $\lambda \in [0, 1]$, $z_\lambda := \lambda z + (1 - \lambda)x_C \in C$ and look at the behavior when $\lambda \xrightarrow{\lambda > 0} 0$.

$$\begin{aligned} 0 &\leq -2\langle x - x_C, z_\lambda - x_C \rangle + \|z_\lambda - x_C\|^2 \\ &= -2\langle x - x_C, \lambda(z - x_C) \rangle + \|\lambda(z - x_C)\|^2 \end{aligned} \tag{3}$$

Dividing by $\lambda > 0$ and then taking the limit as $\lambda \rightarrow 0$ yields the inequality

$$0 \leq -\langle x - x_C, z - x_C \rangle$$



2/ Fundamental Properties of the Orthogonal Complement

Recall that the **orthogonal complement** of a subset $A \subset H$ is the subset

$$A^\perp := \{y \in H \mid \forall x \in A, \langle x, y \rangle = 0\}$$

Proposition (p. 407 (2008 ed.) or p. 427 (2020 edition))

- a) For any subset $A \subset H$, $A^\perp$ is a closed linear subspace of H .
- b) If F is a closed linear subspace of H then
 - (i) $H = F \oplus F^\perp$,
 - (ii) $\begin{cases} H \longrightarrow H \\ x \longmapsto x_F \end{cases}$ (F closed convex) is precisely the orthogonal projection
 - map $p_F : \begin{cases} F \oplus F^\perp \longrightarrow H \\ v + w \longmapsto v \end{cases}$
 - (iii) $F = F^{\perp\perp}$.
- c) For any linear subspace F of H : $\overline{F} = F^{\perp\perp}$.
- d) F linear subspace, is dense in H if and only if $F^\perp = \{0\}$.

2/ Question a)

Proof: a) $A^\perp$ vector subspace: $\forall y_1, y_2 \in A^\perp, \forall x \in A, \forall \lambda, \mu \in \mathbb{R},$

$$\langle x, \lambda y_1 + \mu y_2 \rangle = \lambda \langle x, y_1 \rangle + \mu \langle x, y_2 \rangle = 0$$

$A^\perp$ closed: let $(y_n)_{n \in \mathbb{N}} \in (A^\perp)^\mathbb{N}$ be a sequence that converges to y .

$$\forall x \in A, \quad \langle x, y_n \rangle = 0 \quad \text{so} \quad \lim_{n \rightarrow \infty} \langle x, y_n \rangle = \langle x, y \rangle = 0$$

(Continuity of the inner product w.r.t. each variable stems from the Cauchy-Schwarz inequality: $|\langle x, y \rangle - \langle x, y_n \rangle| = |\langle x, y - y_n \rangle| \leq \|x\| \|y - y_n\|$)

An alternative argument, in line with the 3rd section, consists in writing

$$A^\perp = \bigcap_{x \in A} \text{Ker} (y \mapsto \langle x, y \rangle)$$

It is thus a closed subspace, as an intersection of closed subspaces.

2/ Question b)

(i) $F \cap F^\perp = \{0\}$: if indeed a vector x is in the intersection, then

$$\langle x, x \rangle = \|x\|^2 = 0$$

Let us now show that $F + F^\perp = H$. F is a closed vector subspace, so it is closed and convex. By the Hilbert projection thm, $\forall x \in H, \exists! x_F \in F$ s.t.

$$d(x, F) = \|x - x_F\| \quad \text{and} \quad \forall z \in F, \langle z - x_F, x - x_F \rangle \leq 0$$

For a given $z \in F$, since the latter is a vector space, the “symmetric” of $w := 2x_F - z$ of z w.r.t. x_F , i.e. s.t. $\frac{w+z}{2} = x_F$, is also in F so we have both

$$\begin{cases} \langle z - x_F, x - x_F \rangle \leq 0 \\ \langle w - x_F, x - x_F \rangle = \langle x_F - z, x - x_F \rangle \leq 0 \end{cases}$$

hence $\langle z - x_F, x - x_F \rangle = 0$, i.e. $(z - x_F) \perp (x - x_F)$. One concludes that $(x - x_F) \in F^\perp$. To sum up:

$$\forall x \in H, \exists! x_F \in F \text{ and } x_2 := x - x_F \in F^\perp, \quad x = x_F + (x - x_F)$$

2/ Question b) (end)

(ii) The projection on the closed vector space F gives rise to the decomposition $H = F \oplus F^\perp$. Conversely, since the two summands are orthogonal, the projection of $h = f_1 + f_2$ on the first component f_1 does give the element that minimizes the distance from h to F .

Remark: when F is a closed vector subspace, then one has equality in (2) and the extra property that $x \mapsto p_F(x)$ is linear and self-adjoint.

(iii) $F = F^{\perp\perp}$. The first inclusion is obvious: let $x \in F$

$$\forall y \in F^\perp, \quad \langle x, y \rangle = 0 \quad \text{so} \quad x \in (F^\perp)^\perp \quad (4)$$

Conversely, let $x \in F^{\perp\perp}$. It is in particular in $H = F \oplus F^\perp$, so there exists $(x_1, x_2) \in F \times F^\perp$ such that $x = x_1 + x_2$.

$$\langle x, x_2 \rangle = 0 = \langle x_1 + x_2, x_2 \rangle = \|x_2\|^2$$

Hence $x = x_1 + 0 \in F$.

2/ Questions c) and d)

$\overline{F} = F^{\perp\perp}$ Let now F be a linear subspace, not necessarily closed.

We know from a) that $(F^\perp)^\perp$ is closed and from (4), that $F \subset F^{\perp\perp}$. Since the closure $\overline{F}$ is the smallest closed subset containing F , one has $\overline{F} \subseteq F^{\perp\perp}$.

Conversely,

$$F \subseteq \overline{F} \implies \overline{F}^\perp \subseteq F^\perp \implies F^{\perp\perp} \subseteq \overline{F}^{\perp\perp} \stackrel{(iii)}{=} \overline{F}$$

d) Since $F^\perp$ is a closed subspace,

$$H = F^\perp \oplus F^{\perp\perp} \stackrel{c)}{=} F^\perp \oplus \overline{F}$$

So F is dense, i.e. $\overline{F} = H$ if and only if $F^\perp = \{0\}$.

A thought exercise: take a minute to see how the hypothesis $F^\perp = \{0\}$ allows us to conclude that for any $h \in H$, there exists a sequence $(x_n)_{n \in \mathbb{N}}$ in F that converges to h .

3/ a) The Riesz Representation theorem

The **topological dual** H' of H is the vector space of continuous linear forms/linear functionals on H .

Theorem (Riesz, p. 407-408 (2008 ed.); p. 427-428 (2020 ed.))

$$\Phi : \begin{cases} H \longrightarrow H' \\ a \longmapsto (y \mapsto \langle a, y \rangle) \end{cases} \quad \text{is a linear bijection.}$$

Proof: to begin with, for any $a \in H$, $\varphi_a := \begin{cases} H \longrightarrow \mathbb{R} \\ y \longmapsto \langle a, y \rangle \end{cases}$ is continuous

(by Cauchy-Schwarz) and linear.

Moreover, $a \longmapsto \varphi_a$ is linear by linearity of the inner product w.r.t. its first variable. It is injective: indeed if $\varphi_a = 0$ then

$$0 = \varphi_a(a) = \langle a, a \rangle = \|a\|^2$$

so $a = 0$.

Φ surjective: Let $L \in H'$. If $L = 0$ then $L = \varphi_0$. Otherwise $\text{Ker}(L)$ is a proper subset of H . It is closed as the pre-image $L^{-1}(\{0\})$ of a closed subset.

Hence by 2/b) (i), $H = \text{Ker } L \oplus (\text{Ker } L)^\perp$ with $(\text{Ker } L)^\perp \neq \{0\}$. In particular, there exists $a \in (\text{Ker } L)^\perp$, $L(a) \neq 0$.

Let us show that $(\text{Ker } L)^\perp = \text{Span}(a)$. $\boxed{\subseteq}$ Let $v \in (\text{Ker } L)^\perp$. Set

$$w := v - \frac{L(v)}{L(a)} a \quad (\text{as in the Gaussian elimination})$$

It is such that $L(w) = L(v) - \frac{L(v)}{L(a)} L(a) = 0$, i.e. $w \in \text{Ker } L$ but it is also in $(\text{Ker } L)^\perp$ as a sum of two elements in that subspace. Hence

$$w \in \text{Ker } L \cap (\text{Ker } L)^\perp = \{0\} \quad \text{and thus} \quad v = \frac{L(v)}{L(a)} a \in \text{Span}(a).$$

Set $b := \frac{L(a)}{\|a\|^2} a \in (\text{Ker } L)^\perp$ and let us show that $L = \varphi_b$.

$$\left\{ \begin{array}{l} \forall x \in \text{Ker } L, \quad \varphi_b(x) = \left\langle \frac{L(a)}{\|a\|^2} a, x \right\rangle = 0 \quad \text{since } a \in (\text{Ker } L)^\perp \\ \forall x \in (\text{Ker } L)^\perp, \quad \exists \lambda \in \mathbb{R}, \quad x = \lambda a \quad \text{so } \varphi_b(x) = \left\langle \frac{L(a)}{\|a\|^2} a, \lambda a \right\rangle = L(\lambda a) \end{array} \right.$$

3/ b) Application: Adjoint of an Endomorphism

Existence, Uniqueness, and Norm of the Adjoint

Let $u \in \mathcal{B}(H)$ (continuous endomorphism).

Show that there exists a unique $u^* \in \mathcal{B}(H)$ (the adjoint of u) s.t.

$$\forall x, y \in H, \quad \langle u(x), y \rangle = \langle x, u^*(y) \rangle. \quad (5)$$

Show that $\|u^*\| = \|u\|$.

Let $y \in H$. The map $x \mapsto \langle u(x), y \rangle$ is a continuous linear functional on H . By the Riesz representation theorem, there exists a unique vector $a_y \in H$ such that

$$\forall x \in H, \quad \langle u(x), y \rangle = \langle x, a_y \rangle. \quad (6)$$

Let us show that $u^* : \begin{cases} H \rightarrow H \\ y \mapsto a_y \end{cases}$ is linear: $\forall (y_1, y_2) \in H^2, \forall (\lambda, \mu) \in \mathbb{R}^2$,

$$\begin{aligned} \forall x \in H, \quad & \langle u(x), \lambda y_1 + \mu y_2 \rangle = \langle x, a_{\lambda y_1 + \mu y_2} \rangle \\ & = \lambda \langle u(x), y_1 \rangle + \mu \langle u(x), y_2 \rangle = \lambda \langle x, a_{y_1} \rangle + \mu \langle x, a_{y_2} \rangle = \langle x, \lambda a_{y_1} + \mu a_{y_2} \rangle \end{aligned}$$

Unicity of a_y in (6) allows to conclude that

$$u^*(\lambda y_1 + \mu y_2) = a_{\lambda y_1 + \mu y_2} = \lambda a_{y_1} + \mu a_{y_2} = \lambda u^*(y_1) + \mu u^*(y_2)$$

Let us now show u^* is bounded/continuous:

$$\begin{aligned} \forall y \in H, \quad \|u^*(y)\|^2 &= \langle u^*(y), u^*(y) \rangle = \langle u(u^*(y)), y \rangle \\ &\leq \|u(u^*(y))\| \|y\| \leq \|u\| \|u^*(y)\| \|y\| \end{aligned}$$

so $\|u^*(y)\| \leq \|u\| \|y\|$ and hence $\|u^*\| \leq \|u\|$.

Notice now that (5) is “symmetric”, i.e. $(u^*)^* = u$. Hence, the previous inequality yields

$$\|u^{**}\| = \|u\| \leq \|u^*\|$$

we thus have the equality $\|u^*\| = \|u\|$.

4/ a) Orthogonal Series and Bessel's Inequality

Decomposition and Projection, (p.408 (2008 ed.), p.428 (2020 ed.))

Let $(e_n)_{n \in \mathbb{N}}$ be an orthonormal sequence of vectors in H . For any $v \in H$, let $\lambda_n := \langle e_n, v \rangle$.

Show that the series $\sum_n \lambda_n e_n$ converges. Denote $w \in H$ its limit.

Show that w is the orthogonal projection of v onto $E := \overline{\text{Span}(e_n)_{n \in \mathbb{N}}}$.

Show **Bessel's inequality**: $\|w\|^2 = \sum_{n=0}^{+\infty} \lambda_n^2 \leq \|v\|^2$.

Proof : We'll show all the assertions concomitantly.

Let $E_n := \text{Span}(e_k)_{k=0}^n$. It is closed as a finite dimensional subspace. Let

us show that $p_{E_n}(v) = \sum_{k=0}^n \lambda_k e_k$: we have seen that $(v - p_{E_n}(v)) \perp E_n$.

In particular: $\forall k \in \llbracket 0, n \rrbracket$, $\langle v - p_{E_n}(v), e_k \rangle = 0 = \langle v, e_k \rangle - \langle p_{E_n}(v), e_k \rangle$.

(If $p_{E_n}(v) = \sum_{k=0}^n \mu_k e_k$ then $\langle p_{E_n}(v), e_k \rangle = \mu_k = \langle v, e_k \rangle = \lambda_k$.)

It now follows from the Pythagorean theorem that

$$\begin{cases} \|p_{E_n}(v)\|^2 = \sum_{k=0}^n \|\lambda_k e_k\|^2 = \sum_{k=0}^n \lambda_k^2 \\ \|v\|^2 = \|v - p_{E_n}(v)\|^2 + \|p_{E_n}(v)\|^2 \geq \|p_{E_n}(v)\|^2 \end{cases}$$

The increasing sequence $(\sum_{k=0}^n \lambda_k^2)_{n \in \mathbb{N}}$ of partial sums is bounded, so it is convergent, and thus a Cauchy sequence. Hence so is $(p_{E_n}(v))_{n \in \mathbb{N}}$:

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall p, q \geq N, \|p_{E_p}(v) - p_{E_q}(v)\| = \sqrt{\sum_{k=p+1}^q \lambda_k^2} < \epsilon$$

Denote w the limit. Taking the limit in the above inequality, yields the

$$\textbf{Bessel inequality : } \lim_{n \rightarrow \infty} \|p_{E_n}(v)\|^2 = \|w\|^2 \leq \|v\|^2$$

Let us finally show that $(v - w) \perp E$ ⁽²⁾: $\forall n \in \mathbb{N}$,

$$\langle v - w, e_n \rangle = \langle v, e_n \rangle - \langle w, e_n \rangle = \lambda_n - \lambda_n = 0.$$

²from which follows that $d(v, E) = \|v - w\|$: (cf. also (2) $\Rightarrow$ (1) in section 1).

$$\forall z \in E, d(v, z) = \sqrt{\|v - w\|^2 + \|z - w\|^2} \geq d(v, E).$$



4/ b) Separable Spaces and Parseval's Identity

A Hilbert space H is **separable** if there exists an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$ such that its linear span is dense in H , i.e.

$$\overline{\text{Span}(e_n)_{n \in \mathbb{N}}} = H$$

Corollary

If H is separable, then for all $v \in H$:

$$v = \sum_{n=0}^{+\infty} \langle e_n, v \rangle e_n \quad \text{and} \quad \|v\|^2 = \sum_{n=0}^{+\infty} |\langle v, e_n \rangle|^2 \quad (\text{Parseval's Identity})$$

Proof: Here $E = H$ so $v = p_E(v) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \lambda_k e_k$.

Such a sequence is a **Hilbert basis**, or **orthonormal basis** of H in the french literature.

